

HDOC – 27th day

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Problem Statement:

Write a C program that displays the following menu and uses a switch statement to print the “Pyramid/Diamond pattern “selected by the user. Use nested loops without arrays. Each pattern may be implemented in a separate parameter less function with no return value.

Aim:

To write a menu driven C program using a switch statement and nested loop to display different pyramids and diamond patterns according to user’s choice.

Algorithm:

- 1) Start
- 2) Display the menu.
 - 2.1) Full pyramid.
 - 2.2) Inverted pyramid.
 - 2.3) Diamond pattern.
 - 2.4) Palindrome pyramid.
 - 2.5) Repeated pyramid.
- 3) Input the user choice.
- 4) Use a switch case statement:
 - 4.1) If choice is 1: Enter the number of rows. Repeat the process for each row. Print the required spaces first. Print odd numbers of stars in each row starting from 1. Move to the next line.

4.2) If choice is 2: Enter the number of rows. Start from the maximum number of stars. Print the required spaces. Reduce the number of stars by 2 in every next row. Continue until only one star is printed.

4.3) If choice is 3: Enter the number of rows. First print a full pyramid up to the middle row. Then print an inverted pyramid below it without repeating the middle row. This forms the complete diamond.

4.4) If choice is 4: Enter the number of rows. For each row, first print numbers in increasing order from 1 to the row number. Then print numbers in decreasing order from one less than the row number back to 1. Move to the next row.

4.5) If choice is 5: Enter the number of rows. For each row, print the row number repeatedly. Print it an odd number of times: 1 time in the first row, 3 times in the second row, 5 times in the third row, and so on.

5) If the user enters a choice other than 1 to 5, display "Invalid choice".

6) Stop

Test Case Table:

Test Case	Choice	Input	Expected output	Actual output	Result
1.	1	4	4 rows containing 1,3, 5, 7 stars.	4 rows containing 1,3, 5, 7 stars.	Pass
2.	2	4	4 rows containing 7, 5, 3, 1 stars	4 rows containing 7, 5, 3, 1 stars	Pass
3.	3	5	9 rows containing 1, 3, 5, 7, 9, 7, 5, 3, 1 stars	9 rows containing 1, 3, 5, 7, 9, 7, 5, 3, 1 stars	Pass
4.	4	5	5 rows; last row: 1 2 3 4 5 4 3 2 1	5 rows; last row: 1 2 3 4 5 4 3 2 1	Pass
5.	5	6	6 rows containing 1, 3, 5, 7, 9, 11 repeated numbers	6 rows containing 1, 3, 5, 7, 9, 11 repeated numbers	Pass
6.	6	—	Invalid choice	Invalid choice	Pass

C Program:

The image shows a screenshot of a terminal window running a C program. The window title is 'kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox'. The menu bar includes 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. Below the menu bar is a toolbar with icons for file operations. The file name 'hdoc27.c' is displayed in the top right corner of the terminal area, along with the path '~/.Desktop/HDOC'. The code is as follows:

```
1 #include<stdio.h>
2 int main()
3 {
4     int choice;
5     printf("___MENU___\n");
6     printf("1:Full pyamid\n\
7     2: Inverted pyramid\n\
8     3: Diamond pattern\n\
9     4: Palindrome pyramid\n\
10    5: Repeated pyramid");
11
12    printf("\nEnter your choice: ");
13    scanf("%d", &choice);
14
15    switch(choice)
16    {
17        case 1:
18            int n, i , j;
19            printf("Enter the number or rows:");
20            scanf("%d", &n);
21            for(i = 1; i <= n; i++)
22            {
23                for(j = 1; j <= n-i; j++)
24                    printf(" ");
25
26                for(j = 1; j <= 2*i-1; j++)
27                    printf("* ");
28
29                printf("\n");
30            }
31
32            break;
33
34        case 2:
35        {
36            int n, i, j;
```

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

hdoc27.c
~/Desktop/HDOC

```
35 {  
36     int n, i, j;  
37  
38     printf("Enter the number of rows: ");  
39     scanf("%d", &n);  
40  
41     for(i = n; i >= 1; i--)  
42     {  
43         for(j = 1; j <= n-i; j++)  
44             printf(" ");  
45  
46         for(j = 1; j <= 2*i-1; j++)  
47             printf("* ");  
48  
49         printf("\n");  
50     }  
51  
52     break;  
53 }  
54  
55 case 3:  
56 {  
57     int n, i, j;  
58     printf("Enter the number of rows: ");  
59     scanf("%d", &n);  
60  
61  
62     for(i = 1; i <= n; i++)  
63     {  
64         for(j = 1; j <= n-i; j++)  
65             printf(" ");  
66  
67         for(j = 1; j <= 2*i-1; j++)  
68             printf("* ");  
69  
70         printf("\n");
```

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

hdoc27.c
~/Desktop/HDOC

```
69
70     printf("\n");
71 }
72 for(i = n-1; i >= 1; i--)
73 {
74     for(j = 1; j <= n-i; j++)
75         printf(" ");
76
77     for(j = 1; j <= 2*i-1; j++)
78         printf("* ");
79
80     printf("\n");
81 }
82
83 break;
84 }
85
86 case 4:
87 {
88     int n, i, j;
89
90     printf("Enter the number of rows: ");
91     scanf("%d", &n);
92
93     for(i = 1; i <= n; i++)
94     {
95         for(j = 1; j <= n-i; j++)
96             printf(" ");
97
98         for(j = 1; j <= i; j++)
99             printf("%d ", j);
100
101         for(j = i-1; j >= 1; j--)
102             printf("%d ", j);
103
104         printf("\n");
```

C Tab Width:

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

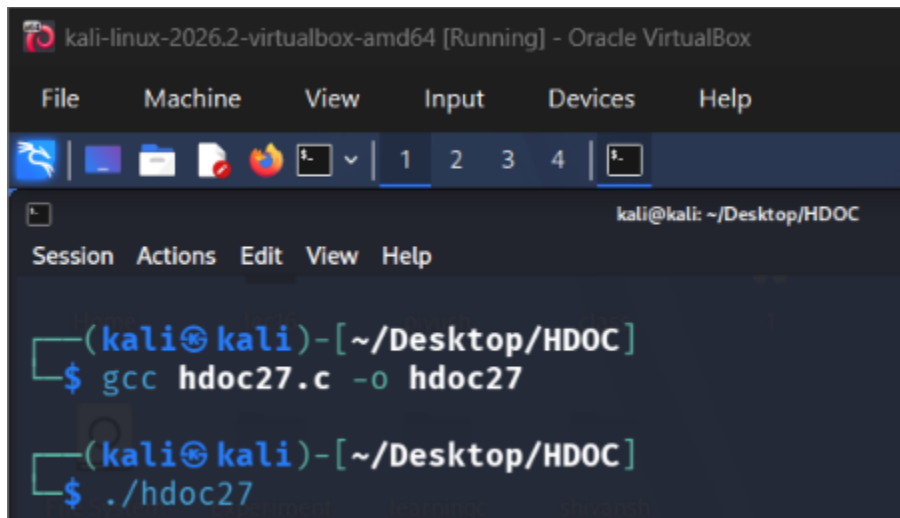
hdoc27.c
~/Desktop/HDOC

```
103
104         printf("\n");
105     }
106
107     break;
108 }
109
110 case 5:
111 {
112     int n, i, j;
113
114     printf("Enter the number of rows: ");
115     scanf("%d", &n);
116
117     for(i = 1; i <= n; i++)
118     {
119         for(j = 1; j <= n-i; j++)
120             printf(" ");
121
122         for(j = 1; j <= 2*i-1; j++)
123             printf("%d ", i);
124
125         printf("\n");
126     }
127
128     break;
129 }
130
131 default:
132     printf("Invalid choice\n");
133 }
134
135 return 0;
136 }
137
138
```

C ▾ Tab

Compile the code using:

```
gcc hdoc27.c -o hdoc27
```



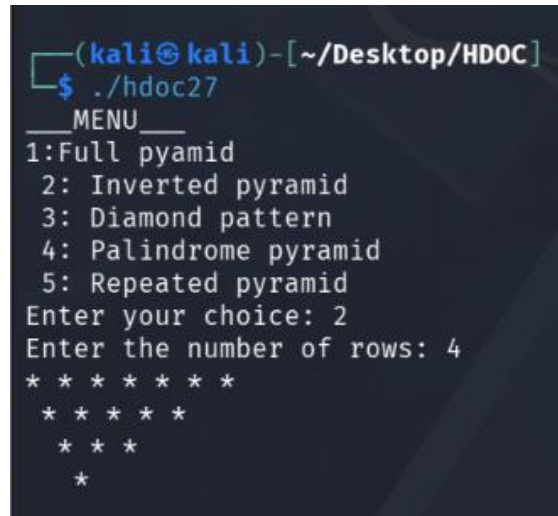
```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~/Desktop/HDOC
Session Actions Edit View Help
(kali@kali)-[~/Desktop/HDOC]
$ gcc hdoc27.c -o hdoc27
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc27
```

Execute the code using:

```
./hdoc27
```



```
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc27
__MENU__
1:Full pyamid
2: Inverted pyramid
3: Diamond pattern
4: Palindrome pyramid
5: Repeated pyramid
Enter your choice: 1
Enter the number or rows:4
*
* * *
* * * * *
* * * * * * *
```



```
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc27
__MENU__
1:Full pyamid
2: Inverted pyramid
3: Diamond pattern
4: Palindrome pyramid
5: Repeated pyramid
Enter your choice: 2
Enter the number of rows: 4
* * * * * * *
* * * * *
* * *
*
```

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle Vir
File Machine View Input Devices
1 2 3 4
kali@
Session Actions Edit View Help
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc27
__MENU__
1:Full pyamid
2: Inverted pyramid
3: Diamond pattern
4: Palindrome pyramid
5: Repeated pyramid
Enter your choice: 3
Enter the number of rows: 5
  *
 * * *
* * * * *
* * * * * * *
* * * * * * * *
 * * * * *
  * * *
   *
```

```
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc27
__MENU__
1:Full pyamid
2: Inverted pyramid
3: Diamond pattern
4: Palindrome pyramid
5: Repeated pyramid
Enter your choice: 5
Enter the number of rows: 6
  1
 2 2 2
3 3 3 3 3
4 4 4 4 4 4 4
5 5 5 5 5 5 5 5
6 6 6 6 6 6 6 6 6 6
```

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle Virtual
File Machine View Input Devices H
1 2 3 4
kali@kali: ~
Session Actions Edit View Help
*
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc27
MENU
1:Full pyamid
2: Inverted pyramid
3: Diamond pattern
4: Palindrome pyramid
5: Repeated pyramid
Enter your choice: 4
Enter the number of rows: 5
1
1 2 1
1 2 3 2 1
1 2 3 4 3 2 1
1 2 3 4 5 4 3 2 1
```

```
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc27
MENU
1:Full pyamid
2: Inverted pyramid
3: Diamond pattern
4: Palindrome pyramid
5: Repeated pyramid
Enter your choice: 6
Invalid choice
```